# Hive Ventures: Enabling a connected food ecosystem from farm to table with Google Cloud

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## About Hive Ventures

Hive Ventures was founded in 2018 with a mission to build a food ecosystem platform that connects food buyers with producers for greater efficiency and sustainability. With in-depth industry knowledge, Hive Ventures empowers suppliers, restaurants, and customers with tailored solutions, including Hive Marketplace for wholesale food sourcing, Hive Restaurant Intelligence (RI) for restaurant management, and Spoonwalk for restaurant reservation, delivery, and reviews.

## Google Cloud results

- Simplifies development of features and services across three apps on microservices architecture
- Complies with global standards to meet customer requirements for data security and privacy
- Simplifies analytics with visualization on Looker Studio

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Industries: Technology

Location: Thailand

(https://cloud.google.com/storage/)), Looker
Studio (https://cloud.google.com/looker-studio)

## About Loxley Orbit

Loxley Orbit PCL is a subsidiary of Loxley PCL., located in Bangkok, Thailand. Loxley Orbit is an official partner of Google Cloud with the expertise in Cloud Native Application, VMs, and Database Migration from old servers onto cloud infrastructure. We provide technology solutions for all businesses. For more than a decade, we are experienced in the IT field to make all work easier for customers. Google Workspace, Google Classroom, Google for Education, Work Transformation, training, and IT consultancy are few of our specialities.

with the expertise in Cloud Native
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Google Cloud (https://cloud.google.com/)

Cloud SQL (https://cloud.google.com/sql/)

Compute Engine (https://cloud.google.com/compute/)

Cloud Storage (https://cloud.google.com/storage/)

Workspace, Google Classroom, Google for
Education, Work Transformation, training,

Thanks to fresh produce and hig
restaurants, Thailand is a food p
locals and tourists alike. Howev
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(Hive), a technology company, w

food crisis by closing the gap be
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Marketplace works like an interc
where suppliers, customers, anc
together to match the wants and needs of the food industry.

Looker Studio (https://cloud.google.com/looker-studio)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Traditionally, farmers rely on middlemen to sell their crops. Without direct market access, farmers lack the bargaining power to secure the best prices. Instead of going through middlemen, farmers can reach a vast network of wholesale and retail buyers on Hive Marketplace. At the same time, restaurants can source fresh produce and other ingredients by comparing prices and product quality from different suppliers.

"By meeting strong privacy standards such as ISO, Google Cloud gives our enterprise customers and consumers the confidence that we're keeping their data safe and protected. Google Cloud provides infinite scalability to handle a large number of transactions, especially for Hive RI."

*—Sittisak Chumpreecha, Vice President of Technology, Hive Ventures*

Hive offers tools and insights powered by Google Cloud (https://cloud.google.com/) to help wholesale suppliers and buyers to operate efficiently and grow their business. Restaurants use Hive Restaurant Intelligence (RI) to streamline business processes, including restaurant promotion and point-of-sale transactions. Hive takes care of the order fulfilment, from sales to delivery, so that farmers can focus on the production.

"By meeting strong privacy standards such as ISO, Google Cloud gives our enterprise customers and consumers the confidence that we're keeping their data safe and protected," says Sittisak Chumpreecha, Vice President of Technology at Hive Ventures. "Google Cloud provides infinite scalability to handle a large number of transactions, especially for Hive RI."

Hive uses Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) (GKE) and Compute Engine (https://cloud.google.com/compute) VMs for its development stack and Cloud SQL (https://cloud.google.com/sql) for PostgreSQL as its relational database.

## Innovating with faster application development on GKE

From day one, Hive built a microservices architecture on GKE for a scalable and secure platform. Addressing scalability is crucial since transactions can spike, particularly when Hive runs digital campaigns to attract new customers. GKE automatically scales for a large number of transactions, so engineers don't need to worry about provisioning or managing servers.

"Google Kubernetes Engine makes it easy to manage code so developers can build and deploy features as independent components," says Chumpreecha. "It gives us the ability to scale resources for a specific microservice for the best performance without scaling the whole application."

Microservices, such as payment and logistics, connect frontend components to backend APIs that deliver business functions on the application. For example, when a customer logs in to the application, the authentication service retrieves the relevant customer profile in the database. When a consumer makes a credit, debit, or QR code payment for their delivery order from a restaurant on Spoonwalk, for example, the payment service communicates with the customer's payment service provider and processes the card payment or bank transfer into the restaurant's account.


*Hive app*

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Hive deploys its 200 microservices to four environments, Production, User Acceptance Testing (UAT), Development, and Sandbox for Quality Assurance (QA). Each microservice has its separate database, giving Hive the flexibility to choose the database technology that's most suitable for each service.

## Transforming the food industry with data on Cloud SQL

Cloud SQL is the workhorse for Hive that processes more than 100,000 transactions every day. Transactions are tied to each customer profile in Cloud SQL so customers can accurately track each order and payment on their dashboard. Since sellers are often unfamiliar with essential business processes for procurement, Hive streamlines recordkeeping by generating business documents such as purchase orders and tax invoices based on their transaction data.

Hive uses MongoDB for its NoSQL database to store attributes that buyers can search such as name and description in its product catalog. When sellers upload a photo of their farm produce, the image is stored in Cloud Storage (https://cloud.google.com/storage) and served through the application when someone clicks on the product.

"When brand owners sell their products in retail stores, they don't know who buys their product. We want to give product owners data insight to compete with retailers. By analyzing big data with Google

Cloud, we help product owners see the big picture and identify opportunities to grow their business."

*—Napapat Pitaksiripan, Co-Founder and Chief Executive Officer, Hive Ventures*

For simple data analysis, teams at Hive use Looker Studio (https://cloud.google.com/looker-studio) as a visualization dashboard to combine data from different sources. For example, the customer support team uses it to track product issues and returns. By analyzing feedback on why customers are sending products back, Hive has plans to use AI for recommendations that help sellers reduce the return rate.

Hive wants to translate its success with its food ecosystem platform into the retail industry, with data as a differentiator. "When brand owners sell their products in retail stores, they don't know who buys their product. We want to give product owners data insight to compete with retailers," says Napapat Pitaksiripan, Co-Founder and Chief Executive Officer at Hive Ventures. "By analyzing big data with Google

Cloud, we help product owners see the big picture and identify opportunities to grow their business."